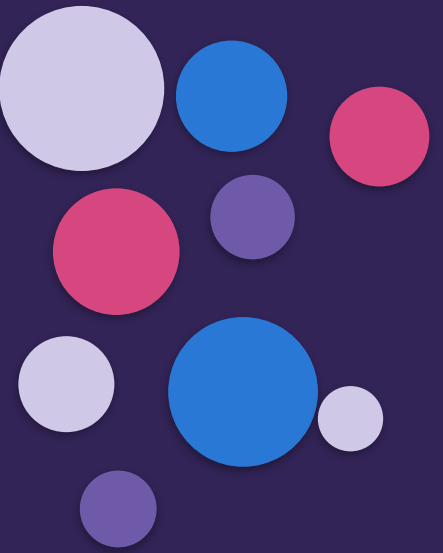
Nuclei Instance Segmentation on PanNuke



Progress report — Course Project 5

October 15, 2026 · bdd26 / jxh

The task

- Find every nucleus in H&E images
- Separate the ones that touch
- Classify each into 1 of 5 types

Dead (dying) nuclei: the hardest class

~7,900 image patches

19 tissue types

5 nucleus classes

How we score: PQ

higher = better · the official PanNuke metric

A prediction only counts when all three are right:

1

Found?
it is detected

2

Boundary right?
shapes overlap

3

Type right?
the class matches

Official 3-fold protocol **Last checkpoint** **No leaked weights**

Reproductions on PanNuke land at mPQ 0.46 – 0.50; every point is hard-won.

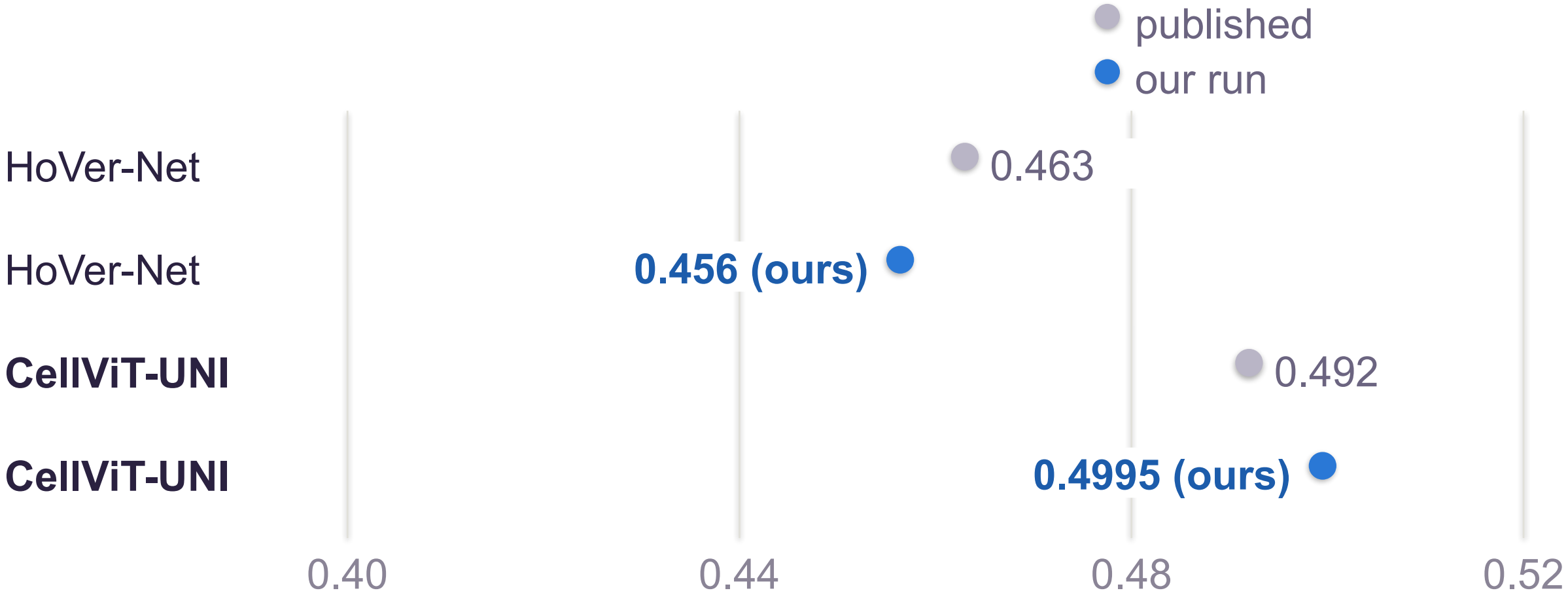
First: a pipeline you can trust

1. **Compact data pipeline —** streamed from zip: 37 GB → 3 GB
2. **Exact evaluator —** bit-identical to the official metric code
3. **One strict stack —** shared evaluator; provenance disclosed
4. **403 tests (prior full run) —** metrics, decoding, targets, plumbing

Own training and released-weight protocols are reported separately.

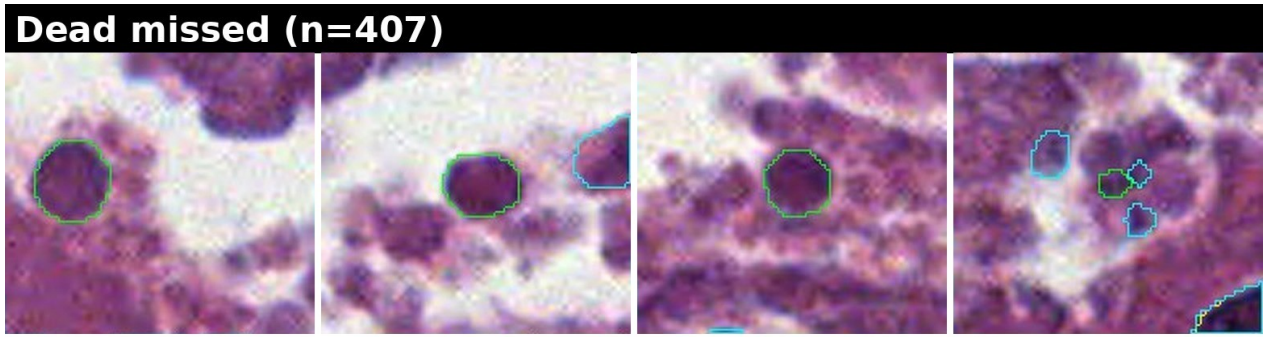
Baselines under the official protocol

mPQ, 3-fold mean, no TTA



Reference numbers are close; checkpoint provenance remains explicit.

A major baseline deficit: missed Dead nuclei



Green: ground-truth Dead nuclei the model missed entirely

38.5%

of Dead nuclei missed entirely (IoU < 0.1)

8 – 15%

missed rate, other classes

69% of matched Dead are typed correctly

→ **a detection-recall deficit**

Stage 1 — three new-paradigm attacks

Sep 25 – 28 · all judged against pre-registered rules

A

Language-anchored typing

CONCH VLM re-types nuclei from text

Null → diagnosis

B

Failure-driven synthesis

diffusion makes hard examples

Closed — no gain

C

Counterfactual stress test

re-render images, keep labels

Typing-only proxy

All three closed with informative negatives — and one key diagnosis.

Stage 1A — language anchoring (CONCH)

the model can name Dead nuclei — it just doesn't find them

0.695
zero-shot typing accuracy (CONCH pooling)

0.88
zero-shot Dead recall

- **Fused into the model:** test mPQ unchanged (0.4951 → 0.4953)
- **Also null:** loss re-weighting · balanced sampling · recovery

Tested re-typing recipes did not repair the detection deficit.

Stage 1B — generative synthesis (PixCell)

make more hard examples with a tissue generator

- **Mined the failures:** missed Dead are small & isolated; merges sit in same-class clusters
- **Copy-paste real Dead nuclei:** hurts — Dead PQ −0.031

- **In-context diffusion (1,146 images):** no gain — mPQ 0.4900, inside seed noise
- **Structural blocker:** mask-only conditioning cannot steer appearance by class

Verdict: closed

Honest scope: these recipes failed — other data interventions stay open.

Stage 1C — counterfactual stress testing

keep the labels, re-render the image → cheap robustness probe

- **Idea:** a generator as an evaluation simulator for stress testing
- **Result:** CF ranking vs real ranking is inverted (ρ = −0.89)
- **Mechanism:** typing rank agrees (ρ +1.00), detection gaps are not reproduced

Only three independent architectures; this renderer is not a transfer proxy.

Verdict: typing stress test, not a transfer proxy

Stage 2 — two attacks on the Dead deficit

Sep 28 – Oct 1 · detection-first architecture vs 2× resolution

Detection-first architecture

KongNet (CenterNet-style)

0.346 mPQ (−11 pts)

- touching same-class nuclei fuse
- interior-Dead miss: 0.50 vs 0.35

Negative

2× working resolution

512 px input & decoding

Dead ↑ at a priced cost

- interior-Dead miss 0.35 → 0.25
- cost: mPQ −0.009 · bPQ −0.007

Priced trade

Resolution helps Dead — can we keep the gain and drop the cost?

Stage 3 — reducing the bPQ trade-off

Oct 2 – 5 · four audited steps

1 Scale-aware decoding
fix decoder constants → recovers ~70% of the cost

2 M/S lever
val-selected merge + fragment fix — 9/9 runs

3 P2: gated additions
borrow 2× detections — Dead rises, guard fails

4 EF-P2: 30px filter
keep credible additions — near-zero bPQ tax

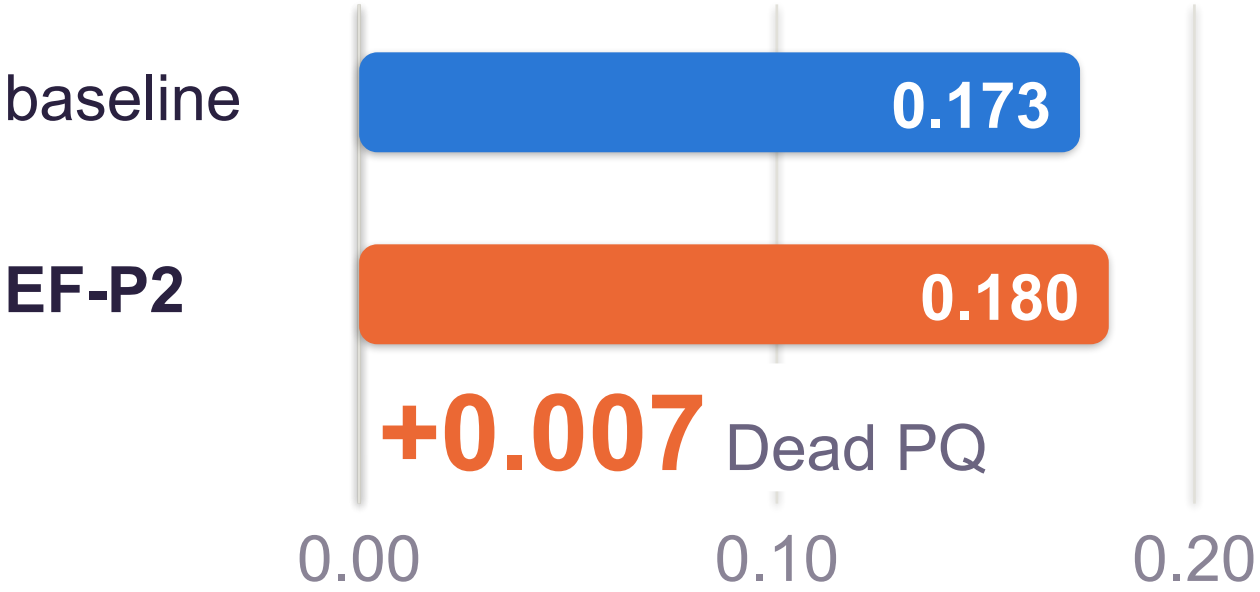
EF-P2 passes the pre-registered rule on 9 matched pairs.

Mean bPQ loss shrank to −.000009; this is not non-inferiority proof.

EF-P2 — a historical Dead-centred candidate

3-split mean over 9 matched (split, seed) pairs

Dead PQ — the target metric



0.4976 → 0.4983
mPQ (+0.0007)

0.6653 → 0.6653
bPQ (Δ ≈ 0 — the guard holds)

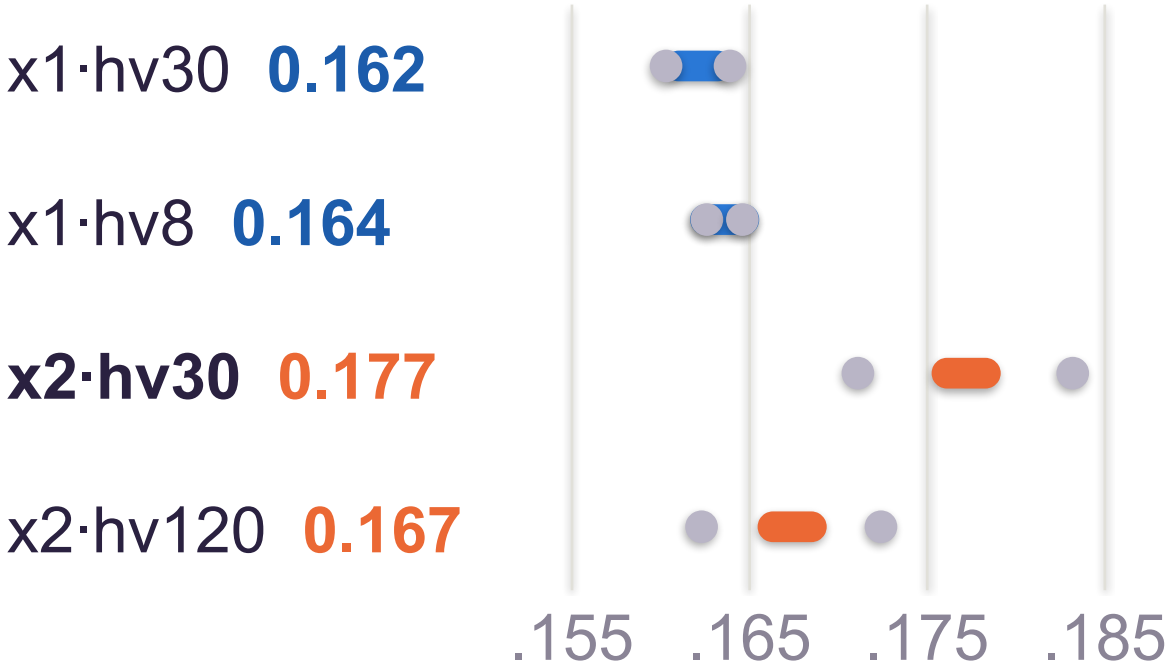
8 ↑ · 1 = · 0 ↓
per-pair Dead PQ signs

Candidate, not proof: CIs span zero · test folds reused across rounds

E2a: scale effects survive matched HV support

E2a · four arms × two seeds · 8/8 runs audited · validation fold only

Dead PQ by arm



gray dots: seeds · bar: arm mean

9.6%

of the 2× gain reproduced by the x1 cutoff change

4 / 4

matched-support pairs: strict Dead lower at 2×

HV threshold stays 30

8/8 runs hash-audited (audit − original = 0) · no test fold read

Dead specialist branch — closed before any training

DSB · four pre-registered gates · stop rule fired Oct 8

A **Base check** — base x1·hv30
gate-lowering explains 9.6% of the 2× gain

B **Conflict premise** — premise refuted
mean cosine +0.749 · 0/64 batches negative

C **Threshold decoupling** — too weak
peak +.0021 — under the +.005 close line

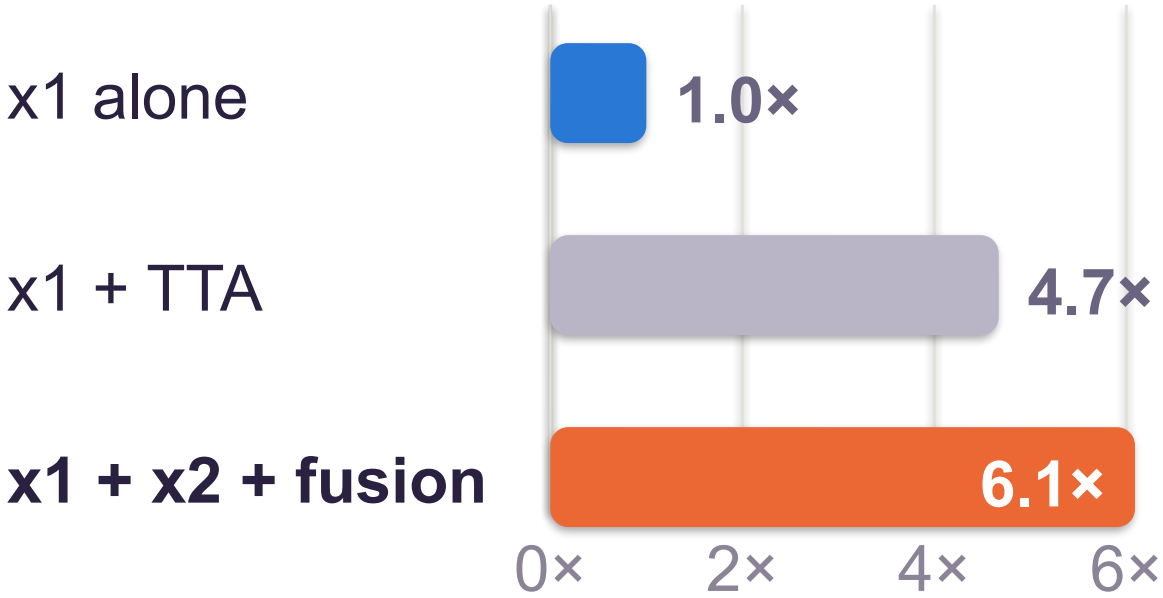
D **Oracle headroom** — headroom confirmed
+.156 Dead PQ if the 267 missed were found

Closed before expert training; gradient probe did use GPU compute.

What the two-model candidate costs

E0 · wall-clock on one shared A100 · unmodified inference path

Cost vs single-model x1



26 ms

per-image CPU fusion · frozen path replayed exactly

1.3×

over plain x1-TTA — which adds no Dead detections

batch 32 · shared GPU · absolute ms load-contaminated, ratios held 6.1–6.3

EF-P2 = mechanism evidence & teacher candidate — not a deployment win (pre-registered E0 rule).

Why trust these numbers

- **Pre-registered rules —** decision criteria fixed before any test data is read
- **Validation-only tuning —** last checkpoint; test folds reused across historical rounds
- **Paired seeds where specified —** seed noise measured: mPQ ±0.002, Dead PQ ±0.007
- **Honest negatives —** failed routes are reported as carefully as wins

Published ‘all-data’ checkpoints are never scored as test results.

Stage 5: typing headroom did not become a method

Oct 9–12 | fixed geometry, train/validation dev probe

GT-assisted oracle: mPQ .4995 → .6632
An upper bound with true labels, not an achievable method score.

T0–T3: frozen-feature read-outs closed
Best official mPQ change +.0002; Dead −.0053 trips the guardrail.

Interpretation
Simple post-hoc typing failed; this does not rule out all joint training.

Source: stage report V; old typing-oracle/T bootstrap CIs are not reused

Label quality: detector agreement, expert review absent

E1-C and E1-T | evidence on existence and typing

Existence: 66–73% vs 0.75% null
Zero-GT-overlap candidates with ≥3 cross-fold / cross-architecture supporters.

Typing: 72.5% base label vs 18.2% GT label
Jury votes on base mistypes; intrinsic flip rate 1.9%.

Evidence boundary
Correlated-detector resonance, not biological confirmation. Human review unavailable.

Source: stage report V / E1-C / E1-T; model pilot validated process only

Candidate utility and merged-label diagnostics

E3 closed at dev; M remains descriptive

E3 utility is predictable: AUC .721 / .737
Learned rows fail the frozen strict-Dead floor; identity fallback; line closed.

M: merging Inf/Conn changes official mPQ by +.0519
Largest response among 10 class pairs; oracle gap shrinks 35.2% in merged space.

Five-class method scores stay unchanged
A merged-label metric response is not recoverable performance or additive label-error loss.

Source: stage report V; RESEARCH_PLAN M entry; M used historical test artifacts

E4: corrected visibility probe passes its three gates

Corrected 2026-10-14 artifacts | single checkpoint, validation fold2 only

H1 area-standardized edge–interior Dead miss gap
+.1738 [.1060, .2528]

H2 difference-in-differences
+.1474 [.1392, .1557]; interpolation control −.0006

H3 fragment-area contrast
+.8175 [.7968, .8393]; does not prove HV zeroing causes missed nuclei

Source: E4 corrected results; old train-mode run and crude H1 gate superseded

Visibility supervision: four controlled dev arms

Seeds 19 / 1 | fold1 train, fold2 validation | 130 epochs, last checkpoint, x1, no TTA

A0: retrained baseline; A1: stronger crop exposure
RandomSizedCrop probability .1 → .5; HV min_size=30 in both.

A2: visible HV for newly truncated fragments
Same crops as A1; min_size=1 only for eligible truncated objects under 30 working pixels.

A3: min_size=1 for every visible object
Actual target difference is limited to objects below 30 working pixels.

Source: frozen visibility-supervision spec; A2−A1 isolates supervision

Dev validation: seed19 completed

Columns: official mPQ / bPQ / Dead PQ / strict Dead PQ | 2,523 images

A0 baseline **0.4840 / 0.6537 / 0.1611 / 0.1175**
A1 crop 0.4767 / 0.6465 / 0.1934 / 0.1254

A2 visible **0.4749 / 0.6474 / 0.1833 / 0.1199**
A3 low cutoff 0.4789 / 0.6469 / 0.1976 / 0.1183

A2 − A0 **-0.0091 / -0.0062 / +0.0222 / +0.0025**
Official Dead rises, while mPQ and bPQ both cross the frozen −.002 floor.

Source: per-run eval_fold2/summary.json | dev scores; no test or 3-split claim

Dev validation: the performance guardrail is violated

8/8 evaluations complete | original-image gate fails | 2026-10-15 06:06 EDT

Seed1 A0	**0.4784**	**0.6500**	**0.1575**	**0.1239**
Seed1 A1	0.4752	0.6450	0.2005	0.1337

Seed1 A2	**0.4750**	**0.6453**	**0.1844**	**0.1195**
Seed1 A3	0.4722	0.6439	0.1800	0.1178

Frozen dev admission cannot pass

Both seeds fail mPQ/bPQ floors; seed1 strict Dead falls .0044.

Snapshot only; no new runs, no menu expansion, no three-split confirmation

Intervention scope and interpretation

Label audit and logged training exposure are distinct measurements

A2 touches 7 Dead objects; A3 touches 70
One deterministic augmentation per fold1 image: 0.93% vs 9.3% of 751 Dead objects.

Actual training: 33,424 / 33,660 eligible exposures
All-class repeated events (seeds19/1); Dead-specific counts were not logged.

Integrate the final report
Separate official / strict scores, diagnostics and expert-review limits.

Source: fold1 audit and 130-epoch logs; exposures are not unique objects

C2: frozen-window mechanism endpoint

116 Dead / 38 images | fixed C2 windows | max-IoU < .1 misses

Seed19: A0 / A1 / A2 / A3 miss rates
0.474 / 0.440 / 0.466 / 0.457

Seed1: A0 / A1 / A2 / A3 miss rates
0.440 / 0.448 / 0.440 / 0.466

A1 minus A2: positive means fewer misses with selective HV
Mean -0.0086; 95% CI [-0.0409, +0.0177]

Paired tissue/image bootstrap on eligible images; 2 fixed seeds; no population claim

Mechanism evidence does not rescue the failed method

Original-image guards remain decisive | diagnostics are descriptive

Original-image interior Dead matched rate: A0 / A1 / A2

0.613 / 0.615 / 0.609

Original-image matched typing, all classes: A0 / A1 / A2

0.856 / 0.855 / 0.854

Decision: no added seeds or three-split confirmation

A2 fails original-image quality floors; C2 results cannot overturn that conjunction.

Full H1, split/merge, typing and paired-object records retained with input/code hashes

PM-Net feasibility: training-label geometry

Fold1 only | 2,656 original patches | no new model training

Query capacity: maximum 276 objects per patch

256 slots overflow by at least 20 objects in one Uterus patch; 512 covers this audit.

A rounded centroid is not always inside its mask

23 / 63,218 centers fall outside (1 / 967 Dead); 116 images contain zero objects.

Before a new model: geometry and cost checks

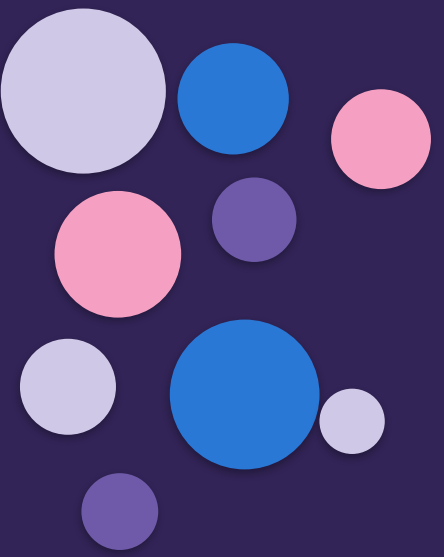
Require legal inside-mask targets, empty-target handling, augmentation and cost audits.

Capacity audit is a lower bound, not a validated query budget or a performance result

Where we stand

- A strict, leak-free pipeline — baselines reproduced.
- Stages I–V complete; E4 supports a controlled dev test.
- EF-P2: +.007 Dead PQ at ~6.1× inference cost.
- **Visibility dev: A2 fails the performance guard.**

Next: integrate the failed dev gate and mechanism limits; integrate final report; human expert review unavailable.



Thank you